DataCORE

Exploring Data with Code

Afternoon: DataCORE

1. **Orientation — Tools & Environment**
2. **Exploring Your First Dataset — NBA Home Advantage**
3. **AI as a Coding Assistant**
4. **Scientific Storytelling — Data Captions**

DataCORE isn't about software — it's about asking questions and finding answers in data.

1:00 – 1:20 · Orientation

DataCORE Orientation

- Everything runs in **Google Colab**, in your browser — **no installation**.
- You'll **open a notebook**, **run a cell**, and see output instantly.
- Today's first cell just says hello →

Big idea: DataCORE is *not* about software. It's about **asking questions and finding answers in data**.

Python R

```python
print("Hello, Research!")
```

```
Hello, Research!
```

1:20 – 2:20 · Exploring Your First Dataset

Step 1 · Load & peek at the data

Open the shared notebook, then **look before you leap** — preview the rows and the shape.

Python R

```
import pandas as pd # load pandas library
df = pd.read_csv("nba_home_away.csv")
print(df.head())        # first 5 rows
```

```
         date      home_team              away_team  home_pts  away_pts  home_wins  away_wins
0  2023-10-24  Atlanta Hawks  Minnesota Timberwolves        83       106          0          1
1  2023-10-24  Atlanta Hawks     Cleveland Cavaliers       147       126          1          0
2  2023-10-24  Atlanta Hawks          Boston Celtics       101       107          0          1
3  2023-10-24  Atlanta Hawks        Sacramento Kings       114       120          0          1
4  2023-10-24  Atlanta Hawks         Houston Rockets       102        99          1          0
```

```
df.shape         # rows = games, cols = fields
```

```
(1230, 7)
```

How many games are in the data? How many teams?

Step 2 · Basic statistics at a glance

Python R

```
print(df.describe().round(2))     # summary stats per column
```

```
       home_pts  away_pts  home_wins  away_wins
count   1230.00   1230.00    1230.00    1230.00
mean     111.54    110.30       0.58       0.42
std       14.72     11.07       0.49       0.49
min       60.00     75.00       0.00       0.00
25%      102.00    103.00       0.00       0.00
50%      112.00    110.00       1.00       0.00
75%      122.00    118.00       1.00       1.00
max      163.00    149.00       1.00       1.00
```

One line gives you the **shape of the whole dataset** — means, spread, and extremes.

Step 3 · Calculate the home win rate

`home_wins` is **1** when the home team won and **0** when it lost — so its **average is the win rate**.

Python R

```python
df["home_wins"].mean()     # share of games won by the home team
```

```
np.float64(0.5829268292682926)
```

Is it above 0.50? By how much? That's the **home advantage** in one number.

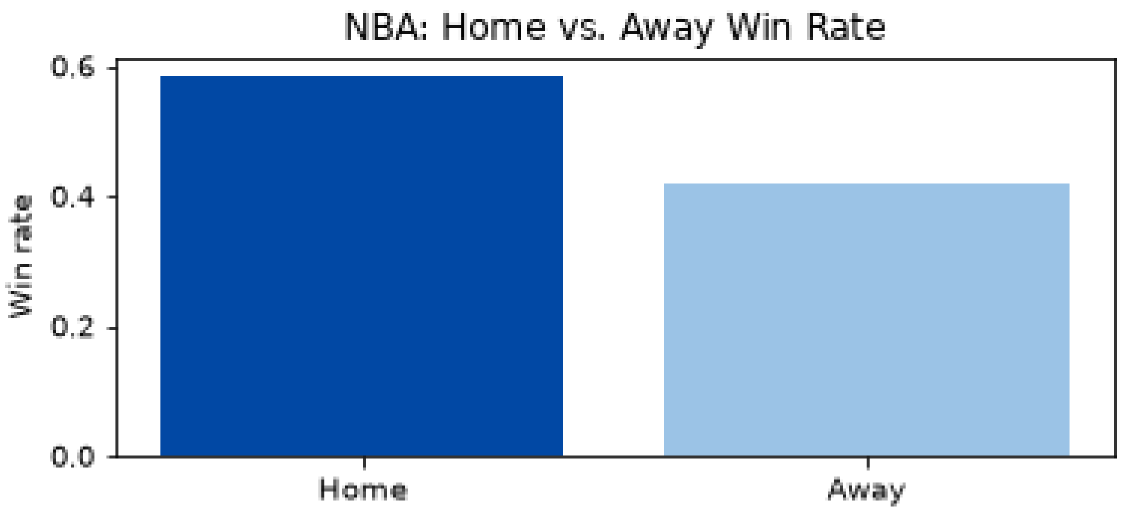
Step 4 · Make a bar chart

Python R

```python
import matplotlib.pyplot as plt

home = df["home_wins"].mean()
away = 1 - home

plt.figure(figsize=(6, 2.4))
plt.bar(["Home", "Away"], [home, away],
        color=["#0148A4", "#9bc3e6"])     # color the bars
plt.ylabel("Win rate")
plt.title("NBA: Home vs. Away Win Rate")  # title & labels
plt.show()
```



🤖 You asked the AI for a "win percentage" column...

It instantly wrote this. You run it — and the numbers look **way too high** (over 100%!). Can you spot the bug?

Python | R

```python
# AI: "what % of games did the home team win?"
home_win_pct = df["home_wins"].sum() / df["away_wins"].sum() * 100    # ÷ away wins
home_win_pct
```

```
np.float64(139.76608187134502)
```

🔍 That gives **139.8%** — home wins should be divided by **total games**, not away wins.

Fix it: divide by total games

Python R

```python
home_win_pct = df["home_wins"].sum() / len(df) * 100
home_win_pct
```

```
np.float64(58.292682926829265)
```

Key lesson: AI *writes* code — **humans verify it**. You have to understand what the code does.

2:45 – 3:00 · Scientific Storytelling

Write a 2-sentence caption for your chart

1. What does the chart **show**?

2. What is **surprising** or notable?

*"Home teams won **58.3%** of games in our dataset, compared to **41.7%** for away teams. The edge is real, but smaller than the roar of a home crowd on TV might suggest."*

Use **precise language** — exact numbers, clear comparison.

Read your caption to your **partner**. Is every number specific and correct?